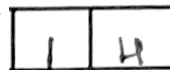




TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Time Allowed : 1.30 Hours]

MATHEMATICS

[Max. Marks : 45

PART - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives **10 x 1 = 10**
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. A circular template has a radius of 10 cm. The measurement of radius has an approximate error of 0.02 cm. Then the percentage error in calculating area of this template is
 (1) 0.2% (2) 0.4% (3) 0.04% (4) 0.08%
2. If $v(x, y) = \log(e^x + e^y)$, then $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y}$ is equal to
 (1) $e^x + e^y$ (2) $\frac{1}{e^x + e^y}$ (3) 2 (4) 1
3. The approximate change in the volume V of a cube of side x metres caused by increasing the side by 1% is
 (1) $0.3xdm^3$ (2) $0.03xm^3$ (3) $0.03x^2m^3$ (4) $0.03x^3m^3$
4. If $w(x, y, z) = x^2(y - z) + y^2(z - x) + z^2(x - y)$, then $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z}$ is
 (1) $xy + yz + zx$ (2) $x(y + z)$ (3) $y(z + x)$ (4) 0
5. A balloon rises straight up at 10 m/s. An observer is 40 m away from the spot where the balloon left the ground. The rate of change of the balloon's angle of elevation in radian per second when the balloon is 30 metres above the ground.
 (1) $\frac{3}{25}$ radians/sec (2) $\frac{4}{25}$ radians/sec (3) $\frac{1}{5}$ radians/sec (4) $\frac{1}{3}$ radians/sec
6. The abscissa of the point on the curve $f(x) = \sqrt{8 - 2x}$ at which the slope of the tangent is -0.25?
 (1) -8 (2) -4 (3) -2 (4) 0
7. The number given by the Rolle's theorem for the function $x^3 - 3x^2$, $x \in [0, 3]$ is
 (1) 1 (2) $\sqrt{2}$ (3) $\frac{3}{2}$ (4) 2
8. The curve $y = ax^4 + bx^2$ with $ab > 0$
 (1) has no horizontal tangent (2) is concave up (3) is concave down (4) has no points of inflection
9. For the function $f(x) = x + \frac{1}{x}$, $x \in [1, 3]$, the value of c for mean value theorem is
 (1) 1 (2) $\sqrt{3}$ (3) 2 (4) None of these
10. The maximum slope of curve $y = -x^3 + 3x^2 + 9x - 27$ is
 (1) 0 (2) 12 (3) 16 (4) 32

PART - II

1. Answer any 4 questions
2. Each question carries 2 marks
3. Question number 16 is compulsory

4 x 2 = 8

11. Explain why Rolle's theorem is not applicable to $f(x) = x - 2 \log x$, $x \in [2, 7]$ 12. Evaluate the following limits, if necessary use l'Hôpital Rule : $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec x}{\tan x}$ 13. Find the asymptotes of the function $f(x) = \frac{1}{x}$.

14. Let us assume that the shape of a soap bubble is a sphere. Use linear approximation to approximate the increase in the surface area of a soap bubble as its radius increases from 5 cm to 5.2 cm. Also, calculate the percentage error.



15. Find the partial derivatives of $g(x, y) = 3x^2 + y^2 + 5x + 2$, at $(1, -2)$

16. Show that $f(x) = \frac{x}{\sin x}$ is increasing on $\left(0, \frac{\pi}{2}\right)$.

PART - III

1. Answer any 4 questions

2. Each question carries 3 marks

3. Question number 22 is compulsory

17. Write down the Taylor series expansion, of the function $\log x$ about $x=1$ upto three non-zero terms for $x > 0$.

18. Find the intervals of monotonicity and hence find the local extrema for the function $f(x) = x^{\frac{2}{3}}$.

19. Find the local extremum of the function $f(x) = x^4 + 32x$.

20. Assuming $\log_{10} e = 0.4343$, find an approximate value of $\log_{10} 1003$.

21. If $v(x, y) = x^2 - xy + \frac{1}{4}y^2 + 7$, $x, y \in \mathbb{R}$, find the differential dv .

22. The volume of a cube is increasing at a rate of $9\text{cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 10 cm ?

PART - IV

1. Answer all the questions

2. Each question carries 5 marks

23. a) A conical water tank with vertex down of 12 metres height has a radius of 5 metres at the top. If water flows into the tank at a rate 10 cubic m/min , how fast is the depth of the water increases when the water is 8 metres deep?

(OR)

b) If the curves $ax^2 + by^2 = 1$ and $cx^2 + dy^2 = 1$ intersect each other orthogonally then, show that $\frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}$.

24. a) A farmer plans to fence a rectangular pasture adjacent to a river. The pasture must contain $1,80,000\text{ sq.mtrs}$ in order to provide enough grass for herds. No fencing is needed along the river. What is the length of the minimum needed fencing material?

(OR)

b) If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$. Show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$.

25. a) Let $U(x, y) = e^x \sin y$, where $x = st^2$, $y = s^2t$, $s, t \in \mathbb{R}$. Find $\frac{\partial U}{\partial s}$, $\frac{\partial U}{\partial t}$, and evaluate them at $s = t = 1$.

(OR)

b) Find the value of $\sqrt[3]{1.02} + \sqrt[4]{1.02}$ by linear approximation.